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Control Tower UI/UX & Demo Design

Design system, screen-by-screen layout, shared components & pitch-stage demo UX

Document: 3 of 3 — UI/UX Plan

Companion documents: Business Plan & Demo Narrative · Technical Plan

Audience: Founder, building solo with Claude Code (design brief — no code in this document)

Date: November 2026

1. Design Principles for a Pitch-Demo UI

This is not an enterprise SaaS UI being designed for daily power users — it is a UI being designed to be convincing in a ten-minute pitch, on a projector or a shared screen, to someone evaluating whether to invest or run a pilot. That changes the priorities from a normal product design brief:

- **Clarity over completeness.** Every screen should be readable at a glance from across a room; dense power-user tables can wait for Phase 1.
- **The AI should visibly explain itself.** Every flag, score or recommendation needs a one-click "why" — this is the single highest-leverage UI pattern in the whole product (ties to Business Plan §10 and Technical Plan §8).
- **One consistent "mission control" aesthetic.** A dark, operations-center feel (navy base, teal/cyan AI accent, amber/red severity) ties all seven modules together visually, reinforcing the "one control tower" story even before anyone reads a word.
- **A dollar number on screen, often.** Cost saved, value recovered, revenue at risk — investors remember numbers; make sure most screens surface at least one, clearly labeled as sample-data illustrative (Business Plan §10).
- **Demo-safe by design.** Big-screen legibility, a guided walkthrough, and a one-click data reset (Technical Plan §4) are treated as first-class UI requirements, not afterthoughts.

2. Design System

Element	Specification
Base palette	Navy/charcoal background (#0f2a3f family) for chrome and headers; off-white/light-grey content surface for data-dense screens so tables stay readable.
Accent — "AI"	Teal/cyan (#5fe3cf family) reserved specifically for anything AI-generated: flags, recommendations, scores, explainability panels. Using this color consistently trains the viewer to recognize "the AI said this" within the first minute of the demo.
Severity system	Red = critical/urgent, amber = warning/watch, green = healthy/resolved — applied identically across all seven modules (a red badge means the same urgency in Reconciliation as it does in OOS Prediction).
Typography	A single clean grotesque/humanist sans (e.g. Inter) for all UI text; numbers in a tabular-figure variant so KPI tiles and tables align cleanly.
Component library	Tailwind CSS + shadcn/ui as the base kit, lucide-react for icons, Recharts or Tremor for all charts — one library per concern, used identically in every module (Technical Plan §2).
Motion	Minimal and purposeful only: a soft pulse on a newly-created AlertFlag, a count-up animation on KPI numbers when a screen first loads. No decorative motion.
Iconography	One fixed icon per module, used in the sidebar, the module's own header, and on every AlertFlag card that originates from it — this is how a viewer tells modules apart at a glance in the shared alerts feed.

3. Information Architecture & Navigation

GLOBAL SHELL (PRESENT ON EVERY SCREEN)

Left sidebar: Control Tower Home (always first, always the default landing screen) followed by the seven modules in this fixed order: 3PL Control, Replenishment & Planning, Document AI, Stock Transfer Optimization, Salvage Optimization, Out-of-Stock Prediction, Reconciliation. **Top bar:** global search (optional, demo-nice-to-have), an AI Alerts bell with live count, a persistent "SAMPLE DATA" badge (Technical Plan §4), and a "Reset Demo Data" control. **Presenter-only layer:** a hidden keyboard-shortcut-triggered overlay with presenter notes and a "jump to demo beat" menu matching Business Plan §5's scripted walkthrough.

4. Screen-by-Screen Layout

4.1 Control Tower Home

LAYOUT

Top: a KPI row (4 tiles — e.g. open AI alerts, $ at risk, $ recoverable, on-time %). Center-left: the live **AI Alerts feed**, one card per flag, severity-colored, each showing module icon, one-line explanation, $ impact and a "view" button that deep-links into the originating module. Center-right: a small network/location map showing all 6 demo locations with a status dot per location (healthy / has an open alert). Bottom: a row of module shortcut tiles for the remaining navigation.

Key interaction: clicking any alert card jumps straight into the right module at the right entity — this single interaction is what makes "one control tower" feel real rather than asserted.

4.2 3PL Control

LAYOUT

Carrier scorecard as a row of cards (cost/shipment, on-time %, claims rate, trend arrow); below, a comparison table for an upcoming shipment with a highlighted "recommended" row; an SLA heatmap (carrier × week) for the trailing quarter; a small map showing active shipments in transit.

4.3 Replenishment & Planning

LAYOUT

Left: SKU list with a reorder-urgency indicator per row. Right (on row select): demand forecast chart (history + forecast line + confidence band), a service-level slider that live-updates the suggested order quantity below it, and a "Create Suggested PO" button.

Key interaction: dragging the service-level slider and watching the recommended quantity and the forecast chart's safety-stock line move together — a strong, tactile "it's really computing this" demo beat.

4.4 Document AI

LAYOUT

Left: document inbox (thumbnail + type badge: PO / Invoice / BOL). Center: the source document preview. Right: extracted fields panel, each field color-highlighted by confidence (green = high, amber = needs review), with an "exceptions" sub-list for anything below threshold.

4.5 Stock Transfer Optimization

LAYOUT

Top: a network view (nodes = the 6 locations, sized/colored by surplus or deficit for the SKU currently selected). Below: a ranked list of recommended transfers, each row showing from → to, quantity, estimated cost, and $ saved vs. a new purchase order, with a one-click "Approve Transfer" action.

Key interaction: approving a transfer here should visibly clear the matching alert on the Out-of-Stock Prediction screen and the Control Tower home feed within the same session — the clearest possible proof of the shared data spine.

4.6 Salvage / Liquidation Optimization

LAYOUT

A ranked card list of at-risk SKUs (age, days-of-supply, score), each card expandable to show the recommended disposition channel and an estimated recovered-value number with an "assumption" info-icon tooltip (Technical Plan §5.5). A small donut/bar chart summarizing recommended channel mix across all flagged SKUs.

4.7 Out-of-Stock Prediction

LAYOUT

A risk-tiered SKU table (red/amber/green chips) sorted by days-to-stockout ascending. Selecting a row opens a demand-vs-supply chart and the **explainability panel**: three plain-language lines (e.g. "Sell-through is 18% above trailing average," "No incoming PO before stockout date," "A covering transfer is available at Warehouse C — see Stock Transfer").

Key interaction: the explainability panel's third line links directly into Stock Transfer Optimization — the second cross-module proof point in the demo.

4.8 Reconciliation with AI Flagging

LAYOUT

A 3-way match table (PO | Goods Receipt | Invoice columns) with a status chip per row (matched / warning / critical). Selecting a flagged row opens a diff view highlighting exactly which field disagrees (e.g. unit price $4.20 on the PO vs. $4.65 on the invoice) with the variance and $ impact called out.

5. Shared UI Components

Component	Used in	Notes
AI Flag card	Home feed + every module	The single most important component in the system — must look and behave identically everywhere so trust transfers across modules.
KPI tile	Home + every module header	Number, label, trend arrow, optional "sample data" tooltip.
Severity badge	Everywhere	Red/amber/green, fixed meaning (Section 2).
Explainability panel	OOS Prediction, Salvage, Reconciliation, Stock Transfer	Plain-language "why" text, never a raw model score.
Location/network map	Home, 3PL, Stock Transfer	Shared map component with mode-specific overlays (shipments vs. surplus/deficit).
Guided-tour tooltip	All screens, demo mode only	Section 7.

6. Demo-Mode UX Specifics

Feature	Purpose
Guided walkthrough overlay	Step-by-step tooltips following the exact beats in Business Plan §5, so the founder (or anyone else demoing it) never has to improvise navigation.
"Day 1 / Day 30" time-travel slider	Optional stretch feature: scrub demand history forward to visibly show a healthy SKU crossing into an OOS risk tier in real time — a strong "it's not just a static mockup" moment.
Big-screen mode	A single toggle that scales up font sizes and simplifies density for projector/shared-screen pitches.
Presenter notes overlay	Visible only on the presenter's own screen/notes view, never on the shared/projected view, listing the talking point for the current screen.
Reset Demo Data	One click returns every module to the exact seeded scenario (Technical Plan §4) between back-to-back pitch meetings.

7. Accessibility & Responsiveness Constraints

- Desktop-first: the demo is run by the founder on a controlled device, so mobile responsiveness is explicitly out of scope for Phase 0 — do not spend build time on it yet.
- Contrast ratios should still meet comfortable readability standards (not strict WCAG AA compliance yet) since investors will be reading screens from a distance, not just up close.
- Full keyboard navigability for the guided walkthrough as a demo-safety net in case a mouse/trackpad misbehaves on stage.

8. Component & Screen Inventory Checklist

Screen	Core components required	Build sprint
Control Tower Home	KPI row, AI Flag feed, location map (status mode)	Sprint 2
Replenishment & Planning	SKU list, forecast chart, service-level slider, PO builder	Sprint 3
Out-of-Stock Prediction	Risk-tiered table, demand-vs-supply chart, explainability panel	Sprint 4
Stock Transfer Optimization	Location map (surplus/deficit mode), ranked transfer list	Sprint 4
Salvage Optimization	Ranked card list, channel-mix chart, assumption tooltip	Sprint 5
Document AI	Document inbox, document preview, extracted-fields panel	Sprint 6
Reconciliation	3-way match table, diff viewer	Sprint 6
3PL Control	Carrier scorecards, SLA heatmap, location map (shipment mode)	Sprint 7
Demo-mode layer	Guided tour, big-screen toggle, presenter notes, reset control	Sprint 8

Sprint numbers match the Technical Plan §9 build sequence — build each module's screen in the same sprint as its underlying logic, not before the data is real enough to lay it out against.

9. Constraints Summary

Constraint	Rationale
One component library across all 7 modules, no exceptions	Visual consistency is what sells "one control tower" instead of "seven features bolted together."
No custom illustration/imagery budget	Icon system + data visualization carries the entire visual language; avoids design-asset time a solo founder doesn't have.
Every AI output ships with a one-click "why"	Non-negotiable per Business Plan §10's explainability commitment.
Desktop-first, no mobile work in Phase 0	Pitch demos run on a controlled laptop/projector, not a phone.
Dark "mission control" chrome + light content surfaces	Balances brand feel against actually being able to read dense tables and charts.

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Companion documents: Business Plan & Demo Narrative (document 1) defines why each module exists, its KPIs and the scripted pitch walkthrough this UI is built to serve; Technical Plan (document 2) defines the data and algorithms each of these screens renders.